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A novel model-based internal state observer of a fuel cell system for electric vehicles using improved Kalman filter approach

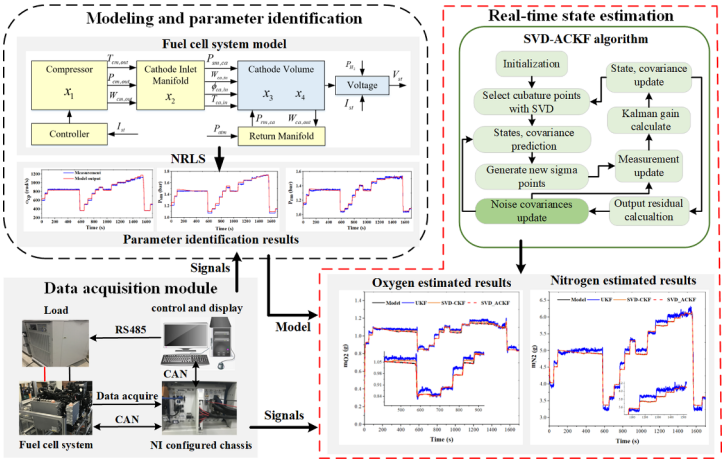
Hao Yuan[a,c], Haifeng Dai[a,b,c,*], Xuezhe Wei[a,c], Pingwen Ming[a,c]

[a] School of Automotive Studies, Tongji University, China
[b] Collaborative Innovation Center for Intelligent New Energy Vehicles, Tongji University, Shanghai, China
[c] National Fuel Cell Vehicle & Powertrain System Research & Engineering Center, No. 4800, Caoan Road, Shanghai 201804, China

HIGHLIGHTS

- A nonlinear state observer based on adaptive cubature Kalman filter is designed.
- Singular value decomposition is employed to improve convergence of observer.
- Compared study for SVD-ACKF, SVD-CKF, and UKF observers are carried.
- Robustness under different initial values and model uncertainties are analyzed.

GRAPHICAL ABSTRACT



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ABSTRACT

Poor air supply system management usually leads to various degrees of oxygen starvation, which affect the durability and reliability of the proton exchange membrane fuel cell (PEMFC). Real-time internal states monitoring of the cathode side is imperative for feedback control to improve the performance of the air supply system and net power. In this paper, a nonlinear internal state observer that estimates the mass of oxygen and nitrogen inside the cathode of a vehicular PEMFC system is proposed. A lumped parameter dynamic system model for the mass transport process on the cathode side is established and parameterized using a nonlinear least-squares algorithm with a working dataset. The model-based nonlinear observer is designed based on an adaptive cubature Kalman filter (ACKF) algorithm. The adaptive adjustment is implemented to match the covariances of process noise and measurement noise in real-time. Furthermore, the singular value decomposition (SVD) is employed to replace the Cholesky decomposition of the covariance of state variables to improve the convergent stability of the iterative calculation process. The performance of the proposed SVD-ACKF observer is validated and compared with the unscented Kalman filter (UKF) observer and SVD based cubature Kalman filter

* Corresponding author at: School of Automotive Studies, Tongji University, China.
E-mail address: tongjidai@tongji.edu.cn (H. Dai).

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(SVD-CKF) observer with no adaptive process. The results show that the SVD-ACKF observer demonstrates the best convergence and accuracy, even though various uncertainties in the initial values of the internal states and system parameters, and upon different power requirements.

1. Introduction

Nowadays, energy shortage and environmental pollution become more conspicuous with the increasing number of vehicles in the world. The development of new energy vehicles will have a significant and far-reaching impact on the pattern of the global automobile and energy industry as well as the social and economic development [1–3]. Proton exchange membrane fuel cell vehicles have become one of the next generations of electric vehicle development targets due to their advantages of fast hydrogenation, long driving range, and zero pollution emissions [4,5]. However, the promotion and application of fuel cell vehicles are limited by the problems of cost life, reliability, and environmental adaptability.

To meet the critical targets of fuel cell vehicles, the subcomponents of the fuel cell system must be controlled and coordinated effectively with high-quality controllers [6]. Existing research reveals that dynamic vehicle conditions could result in tremendous fluctuations in multi-dimensional internal states such as the gas pressure and concentration inside the gas channel, gas diffusion layer, and catalyst layer [7,8]. Insufficient reactant supply could lead to local, even global reactant starvation phenomenon [9]. Eventually, the performance and durability of the fuel cell deteriorate [10,11]. To this end, it is crucial to control reactant concentration at the expected levels to avoid unexpected faults. Promptly modifying the oxygen excess ratio (OER) with improving enough air supply is the most certified approach to prevent oxygen starvation [12]. Nevertheless, the OER is determined by oxygen mass inside the cathode, which can't be measured directly with vehicle-used sensors [13]. Moreover, excessive air supply will cause superfluous air compressor power loss, leading to excessive parasitic power consumption and reducing the efficiency of the PEMFC system [14]. Therefore, if the mass of oxygen and nitrogen within the fuel cell stack could be obtained, such an OER-based air subsystem control can be designed to prohibit starvation and reduce parasitic power loss to improve system efficiency.

As for monitoring methods for reactant concentration, printed circuit board (PCB) current density measurement technology is a popular method, which directly obtains the local characteristics within the PEMFC, and reactant concentration distribution can be assessed [15,16]. Another technique has been demonstrated based on gas sampling analysis, in which the capillary gas sampling tubes are inserted into the gas channel by using a fuel cell with a bipolar plate perforated [17]. In this way, the gas specimen is sampled, and the partial pressure of hydrogen, oxygen, and nitrogen could be obtained in each sampling period using a mass spectrometer. Although these approaches could acquire detail internal reactant mass levels, the PEMFC system is hermetic construction. The requirement for retrofitting fuel cell structure increases the complexity and cost of the stack, which makes it almost impracticable to apply to the vehicular system, so these methods are more fitting for a single cell testing in the laboratory condition. Accordingly, it is necessary to develop a virtual sensor according to the fuel cell system model, and peripheral convinced measurable sensors like the voltage, current, and pressure sensors, to realize the estimation of internal states.

1.1. Literature review

The model is the premise of observer design, and recent efforts published in fuel cell modeling have been thoroughly reviewed in the papers [18,19]. Generally speaking, the research covering the dynamic model for fuel cell observer design is mainly divided into two categories. One is to follow the idea of distributed parameter modeling. The internal state dynamic process and spatial distribution are described by a series of partial differential equations such as continuity equation and Navier-Stokes equation, as well as the conservation equations of macroscopic features (mass, momentum, charge, and energy). The spatial estimation of internal states could be obtained with the distributed parameter model [20,21]. Still, it's time-consuming for the real-time processor of the fuel cell controller and effectiveness needs to be further verified. On the other hand, the lumped parameter model assumes that the spatial distribution of the internal states can be processed as evenly distributed, which significantly reduces the calculation burden of the model. A famous lumped parameter dynamic fuel cell system model developed explicitly for control engineering and observer can be traced back to the work of Pukrushpan et al. [22].

As for model-based observer, most existing techniques for fuel cell internal state estimation can be divided into the following categories: Kalman filter observers, Luenberger observers, sliding mode observers, etc. The Kalman filter is a recursive filter algorithm used to estimate the unknown state of a dynamic system in the sense of minimum mean square error according to the noise and measurement data acquired in discrete real-time periods [23–25]. For the nonlinear systems, the extended Kalman filter (EKF) has been applied by linearizing system dynamics with the present average value and covariance of the states [26,27]. It has been employed in sophisticated observer design techniques with variant equivalent circuit models for state of charge [27–29] and state of health [27,30] estimation of the lithium-ion batteries, and widely applied for state variables and unknown input of electric vehicle driveline [31–34]. As for fuel cell estimation, Pukrushpan et al. [35] proposed an EKF estimator based on an eighth-order linear model to estimate oxygen mass inside the cathode, and the observability of this observer is detected using different measurable variables. Nevertheless, linearization is not suitable for a higher-order and complex PEMFC system model. An alternative approach for nonlinear estimation called unscented Kalman filter (UKF) is further developed to estimate internal states within the fuel cell, including oxygen-depleted mass flow [36,37], internal temperature [38], nitrogen saturation inside anode [39], exhaust valve position [40], and health prognosis [41,42]. Kandepu et al. [43] applied the UKF and EKF observers to determine internal state of solid oxide fuel cells, and the results showed that the UKF observer could converge to the real value faster and more accurate in case of different initial values. Luenberger observer used for the fuel cell has also been well documented in the literature by providing a convergent error dynamic through the feedback gain configuration that the pores of the error dynamics are in the strictly stable region [44–46]. Thawornkuno et al. [47] proposed a Luenberger observer for water content inside the fuel cell, in which the water vapor pressure is calculated from the mass flow balance with measured voltage and the liquid water content is derived from the calculated water pressure and membrane resistance. A comprehensive comparison between Luenberger observer and Kalman filter observer for internal gas pressure estimation is also performed by Nassif et al. [48], and the results show that Kalman filter observer converges faster than Luenberger observer in terms of dynamic response speed. Besides, static feedback gain coefficient through the pole configuration may not satisfy the convergence requirement when dealing with dynamic conditions involve external disturbances. Meanwhile, the state reconstruction process does not consider the influence of noise, so the noise of measured data needs to be filtered before applying for the state estimation with the Luenberger observer. On the other hand, the sliding mode observer has attracted considerable attention since the design of the sliding mode convergence algorithm is independent of the object parameters and disturbances. Hence it guarantees good robustness and

relatively high accuracy. In prophase, first-order sliding mode observer (SMO) was usually chosen for internal state estimation within fuel cell [49–51]. However, there is generally chattering phenomena at the output of the first-order SMO so that the feedback control based on this observer is susceptible to noise and could generate frequent fluctuations on system inputs. Correction techniques like averaging control effect, low-pass filter, and boundary layer design could ease the chattering effects to some extent, but the finite-time convergence and stability characteristics of the observer may be destroyed [52]. Therefore, the high-order sliding mode (HOSM) observers for state estimation within fuel cells raise great attention because of the better performance on mitigating the chattering effect and maintaining the robustness and accuracy as compared with the classic first-order SMO [13,14,20,21], which is favorably utilized for estimation of air excess ratio [53,54], oxygen partial pressure [55,56], hydrogen partial pressure [54,57,58], and internal temperature [58]. But the PEMFC system model is exceedingly complicated and nonlinear, resulting in an exhausting work on the computation of the observation matrix of the HOSM observer.

1.2. Existing challenges and original contributions

Although many studies have been done on the model-based internal state observer for the fuel cell, there are still some critical challenges. First, in the vehicle environment, a wide variety of noise and electromagnetic interference can affect the measured signal profoundly, and the observer should own good robustness against complex perturbed conditions. For example, the Kalman filter observer is very sensitive to process noise and measurement noise. The calculation is hardly to converge if the noise covariances are not appropriately chosen. By comprehensively reviewing state-of-the-art estimation schemes for the observers used for fuel cells, it can be identified that there is a lack of studies on robustness and estimation accuracy improvement considering various uncertainties in the initial values of the internal states and system parameters. Besides, the computational efficiency is essential for vehicular online state estimation, especially for the high-order fuel cell system model-based HOSM observer. However, once the observer model is excessively complex, led to highly nonlinear and state-dependent, it's still far from easy to calculate the observation matrix of the HOSM observer that meets the full rank requirement. The calculation of the EKF observer depends on the Jacobian matrix calculation, which is also quite intractable to implement. Therefore, the design of estimation algorithms to observe the internal critical state should be easily implemented.

Aiming at the problems mentioned above, a nonlinear observer for the mass estimation of oxygen and nitrogen inside the cathode of a vehicular PEMFC system on the basis of an adaptive cubature Kalman filter algorithm (ACKF) is proposed. A noticeable difference from the existing Kalman filter observer is that the cubature Kalman filter observer accomplishes the purpose of higher accuracy and better stability. Apart from this, most of the published articles adopted the system model parameters that directly extracted from the paper [22] or a real-time PEMFC emulator. No papers systematically introduced the process from model parameter identification to observer design and validation using the practical system dataset. In this paper, the parameters of the system model are identified using actual measured data from a 30 kW vehicular PEMFC system rather than directly obtained from other articles, and the process of parameter identification for a real vehicular fuel cell system is introduced. What is more, the estimation accuracy and robustness against uncertainties and measurement noise can be guaranteed because the adaptive adjustment could match the covariances of process noise and measurement noise in real-time. In addition, the singular value decomposition (SVD) instead of Cholesky decomposition is applied to improve the convergence of ACKF. With this improvement, the adaptive process of noise variance is easy to implement. To highlight the performance of the SVD-ACKF observer, the proposed observer is validated and compared with the unscented Kalman filter observer and cubature Kalman filter observer with no adaptive process under a more extended time scale rather than a few seconds. Besides, their robustness under different initial values, model uncertainties, and power requirements are analyzed.

1.3. Outlines of the paper

This paper is organized as follows: Section 2 introduces the PEMFC system used in the experiment and a lumped parameters dynamic system model. The estimation processes with the UKF and SVD-ACKF algorithms based on the system model are presented in Section 3. The

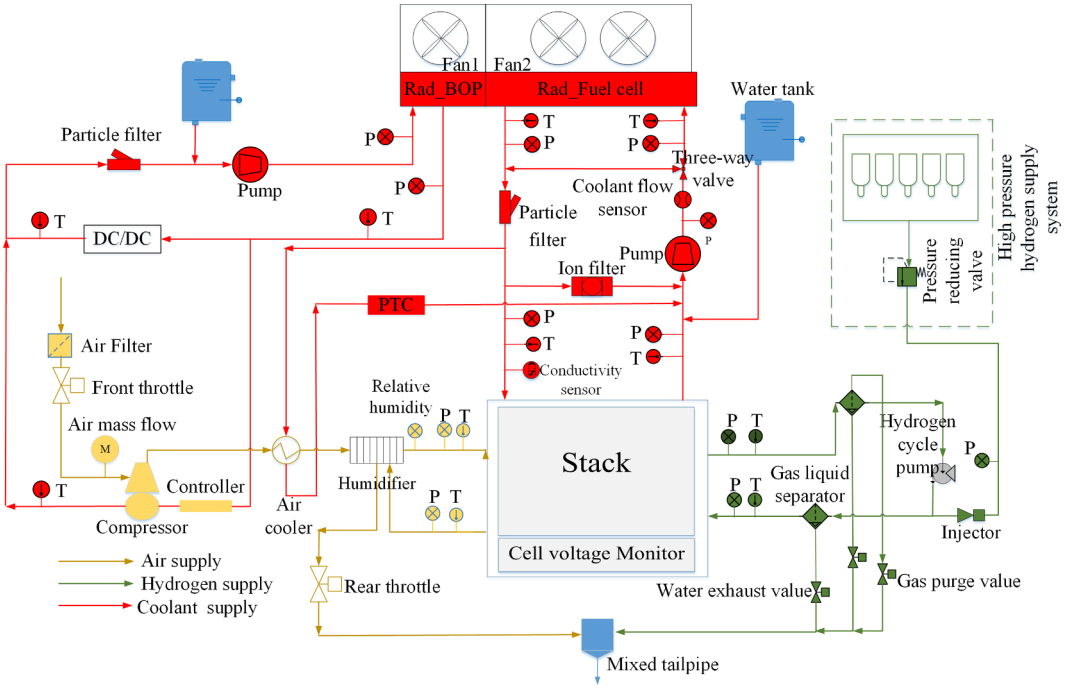


Fig. 1. Schematic diagram of the fuel cell system used in this paper.

implementation of the experiment and validation are described in Section 4. Then, discussion about practical limitations on the application of observers is portrayed in Section 5, followed by concluding summaries and some research lines for future work in Section 6.

2. Nonlinear model of a PEMFC system

2.1. PEMFC system description

The fuel cell system used in this paper is shown in Fig. 1, which is mainly composed of the following four parts: stack, air supply subsystem, hydrogen supply subsystem, as well as thermal management subsystem. The fuel cell stack has a power of 38 kW, which is composed of 130 single cells, and each cell has an effective active area of 340 cm^2. The supplied air is compressed by a compressor to maintain relative high pressure to improve fuel cell performance. In most situations, the air temperature at the outlet of the compressor could reach over 200 °C. To this end, the air must be cooled down by an intercooler to prevent the fuel cell from damaging. The inlet gas is humidified as it passes through a membrane humidifier, in which the opposite side is cathode exhaust gas that contains enough moisture. The inlet air pressure is controlled by adjusting the appropriate angle of the rear throttle with a pulse width modulation voltage signal. The front throttle valve is mainly closed in conjunction with a discharge resistor during the shutdown operating condition to deplete the remaining oxygen on the cathode side, thus to avoid the hydrogen-air interface, which may accelerate the aging of the stack. In the hydrogen supply subsystem, the pressure in the hydrogen storage bottle is usually as high as 30 Mpa and reaches 8–10 bar after passing through the first-stage pressure reducing valve. The pressure on the anode side of the stack usually does not exceed 250 kpa (all pressures in this paper are absolute pressure), and the pressure difference between the cathode and anode should be kept within 50 kpa. Therefore, the injector often adjusts its switching frequency based on cathode pressure feedback so that the pressure difference between the two sides could be controlled within a reasonable range. A hydrogen cycle pump (HCP) is configured to recycle the anode gas from the outlet to the inlet, which could ensure the uniform distribution of hydrogen and improve fuel economy. Moreover, a gas-liquid separator is installed at the outlet of the anode side to avoid HCP and solenoid valve rusting after a long period of operation. Besides, considering condensation due to temperature difference between gas inlet and outlet, a gas-liquid separator is also placed at the inlet ports to reduce the risk of flooding. The thermal management system includes a primary cooling circuit and an auxiliary cooling circuit. Because the auxiliary system generates less heat, the air compressor shares the cooling circuit with the DC / DC, while the cooling circuit of the stack and air cooler is integrated. Meanwhile, the primary cooling comprises a smaller loop bypassing the radiator and cooling fans for warming the stack quickly in the start-up condition and a larger circuit that covers the radiator group for maintaining the stack temperature during regular operation.

2.2. Nonlinear dynamic model

A simplified nonlinear model comprised of the cathode volume and system auxiliary is rebuilt based on the works of Pukrushpan et al. [22]. In principle, the working conditions on the anode side will affect the dynamic characteristics on the cathode side due to the mass transfer process in the membrane, and vice versa. However, inspired by the Ref. [59], we know that this influence is slightly, and anode pressure could be adjusted quickly by the injector so that the model of anode side is ignored in this paper. Besides, stack temperature is assumed constant due to its slow time response of stack temperature.

2.2.1. Air compressor model

A permanent magnet DC motor produced by UQM technologies is utilized to drive the screw compressor. The air compressor model is established based on the rotational parameter model and the mass flow diagram of the compressor [22,60], and the compressor motor current dynamics are neglected due to the shorter electrical time constant compared to the mechanical dynamics [55]. The dynamic characteristic of angular velocity ω_{cp} is expressed as follows:

$$\frac{d\omega_{cp}}{dt} = \frac{1}{J_{cp}}(\tau_{cm} - \tau_{cp}) \tag{1}$$

where J_{cp} is the compressor rotational inertia, τ_{cm} and τ_{cp} are the electromagnetic torque produced by the motor and the compressor load torque, respectively, in which the driving torque is controlled by the polynomial based on the load current. They are denoted as follow:

$$\tau_{cm} = \frac{\eta_{cm}K_t}{R_{cm}}(c_1 I_{st} + c_2 I_{st}^2 + c_3 I_{st}^3 + c_4 I_{st}^4 + c_5 - K_v\omega_{cp}) \tag{2}$$

$$\tau_{cp} = \frac{C_p}{\omega_{cp}}\frac{T_{atm}}{\eta_{cp}}\left[\left(\frac{p_{sm}}{p_{atm}}\right)^{\frac{\gamma-1}{\gamma}} - 1\right]W_{cp} \tag{3}$$

where c_1~c_5 are parameters to be identified, η_{cm} is the compressor motor mechanical efficiency, η_{cp} is the compressor efficiency; K_t, K_v and R_{cm} are the motor constants, C_p is the specific heat capacity of air, γ is the ratio of specific heats of air; p_{sm} and p_{atm} are supply manifold pressure and atmospheric pressure, respectively; W_{cp} is the output mass flow of air compressor, and can be calculated by a nonlinear curve fitting method, which can be expressed as follow:

$$W_{cp} = B_{00} + B_{10}\omega_{cp} + B_{20}\omega_{cp}^2 + B_{01}p_{sm} + B_{11}p_{sm}\omega_{cp} + B_{02}p_{sm}^2 \tag{4}$$

where B_{00}, B_{10}, B_{20}, B_{01}, B_{11}, B_{02} are characteristic fitting parameters of the air compressor.

2.2.2. Air supply manifold model

The cathode supply manifold aggregates the volume of all connected pipes and interfaces between the air compressor and the cathode of the stack. The air dynamics in the supply manifold is usually modeled by the following expression [22,60]:

$$\frac{dp_{sm}}{dt} = \frac{\gamma R_a}{V_{sm}}(T_{cp}W_{cp} - T_{sm}W_{sm,out}) \tag{5}$$

where R_a is the air gas constant, V_{sm} is the volume of supply manifold, T_{cp} and T_{sm} are the air temperature in the compressor and cathode supply manifold respectively. T_{cp} can be expressed by the following equation [60]:

$$T_{cp} = \frac{T_{atm}}{\eta_{cp}}\left(\eta_{cp} + \left(\frac{p_{sm}}{p_{atm}}\right)^{\frac{\gamma-1}{\gamma}} - 1\right) \tag{6}$$

Moreover, because of the small pressure difference between the supply manifold and cathode volume, the outlet mass flow rate $W_{sm,out}$ can be written as a linear nozzle equation [22,55,60]:

$$W_{sm,out} = k_{sm,out}\left(p_{sm} - \frac{R_{O_2}T_{st}m_{O_2}}{V_{ca}} - \frac{R_{N_2}T_{st}m_{N_2}}{V_{ca}} - p_{vap,ca}\right) \tag{7}$$

where $k_{sm,out}$ is the outlet orifice constant of supply manifold; R_{O_2} and R_{N_2} are the oxygen and nitrogen gas constant, respectively; $p_{vap,ca}$ is the vapor partial pressure inside the cathode, V_{ca} is the cathode volume; m_{O_2} and m_{N_2} are the mass of oxygen and nitrogen inside the cathode, respectively.

2.2.3. Cathode volume model

According to the mass conservation, the ideal gas law, the thermodynamic properties of the mixed gas, and the humidity characteristics, a volume sub-model is established to reflect the dynamic gas behavior in the cathode [13,14,22,35,55,60]. The dynamic behavior of gases inside the cathode is given by:

$$\frac{dm_{O_2}}{dt} = W_{O_2,ca,in} - W_{O_2,ca,out} - W_{O_2,react} \tag{8}$$

$$\frac{dm_{N_2}}{dt} = W_{N_2,ca,in} - W_{N_2,ca,out} \tag{9}$$

where $W_{O_2,ca,in}$ and $W_{N_2,ca,in}$ are the input mass flow of oxygen and nitrogen, respectively; $W_{O_2,react}$ and $W_{N_2,ca,out}$ represent the output mass flow of oxygen and nitrogen. $W_{O_2,react}$ is the reaction rate of oxygen, it could be expressed by electrochemical principles:

$$W_{O_2,react} = M_{O_2}\frac{NI_{st}}{4F} \tag{10}$$

where M_{O_2} is molar mass of oxygen, F is the Faraday constant, N is the numbers of single cells, I_{st} is the load current.

Humidification is required before the air enters the stack, and the inlet humidity ratio can be expressed as [60]:

$$\omega_{ca,in} = \frac{M_v}{M_{air}}\frac{\phi_{ca,in}p_{sat}(T_{sm})}{p_{sm} - \phi_{ca,in}p_{sat}(T_{sm})} \tag{11}$$

where $\phi_{ca,in}$ is relative humidity of inlet gas at cathode side, M_v and M_{air} are molar mass of vapor and dry air, $p_{sat}(\cdot)$ is the saturated vapor pressure function. $W_{O_2,ca,in}$ and $W_{N_2,ca,in}$ are denoted as follows:

$$W_{O_2,ca,in} = X_{O_2,ca,in}\frac{1}{1+\omega_{ca,in}}W_{sm,out} \tag{12}$$

$$W_{N_2,ca,in} = (1 - X_{O_2,ca,in})\frac{1}{1+\omega_{ca,in}}W_{sm,out} \tag{13}$$

where $X_{O_2,ca,in}$ is the mass mole fraction of the input air flow, it could be calculated by:

$$X_{O_2,ca,in} = \frac{Y_{O_2,ca,in}M_{O_2}}{Y_{O_2,ca,in}M_{O_2} + (1 - Y_{O_2,ca,in})M_{N_2}} \tag{14}$$

Among them, $Y_{O_2,ca,in}$ is the oxygen molar ratio at cathode inlet. The output flow of the air and nitrogen can be shown as:

$$W_{O_2,ca,out} = X_{O_2,ca,out}\frac{1}{1+\omega_{ca,out}}W_{ca,out} \tag{15}$$

$$W_{N_2,ca,out} = (1 - X_{O_2,ca,out})\frac{1}{1+\omega_{ca,out}}W_{ca,out} \tag{16}$$

where $X_{O_2,ca,out}$ is the mass fraction of the output air flow, $\omega_{ca,out}$ represents the output humidity ratio, $W_{ca,out}$ is the output mixed gas flow, $Y_{O_2,ca,out}$ is the oxygen molar ratio at cathode outlet. Their relationships are calculated as follow:

$$X_{O_2,ca,out} = \frac{Y_{O_2,ca,out}M_{O_2}}{Y_{O_2,ca,out}M_{O_2} + (1 - Y_{O_2,ca,out})M_{N_2}} \tag{17}$$

$$Y_{O_2,ca,out} = \frac{p_{O_2,ca}}{p_{O_2,ca} + p_{N_2,ca}} \tag{18}$$

$$\omega_{ca,out} = \frac{M_v}{Y_{O_2,ca,out}M_{O_2} + (1 - Y_{O_2,ca,out})M_{N_2}} \cdot \frac{p_{vap,ca}}{p_{O_2,ca} + p_{N_2,ca}} \tag{19}$$

in which $p_{O_2,ca}$, $p_{N_2,ca}$ and $p_{vap,ca}$ are partial pressure of oxygen, nitrogen and vapor inside the cathode, they can be calculated by:

$$p_{O_2,ca} = \frac{R_{O_2}T_{st}}{V_{ca}}m_{O_2} \tag{20}$$

$$p_{N_2,ca} = \frac{R_{N_2}T_{st}}{V_{ca}}m_{N_2} \tag{21}$$

$$p_{vap,ca} = \phi_{ca}p_{sat}(T_{st}) = \frac{R_vT_{st}}{V_{ca}}m_{vap} \tag{22}$$

$$W_{ca,out} = k_{ca,out}(p_{O_2,ca} - p_{N_2,ca} - p_{vap,ca}) \tag{23}$$

where T_{st} is the stack temperature, ϕ_{ca} is relative humidity inside the cathode, m_{vap} is the vapor mass inside the cathode and R_v represents the vapor gas constant.

2.2.4. Air return manifold model

Like the supply manifold, the return manifold pressure can be derived by the mass conservation and the ideal gas law [22,60]:

$$\frac{dp_{rm}}{dt} = \frac{R_aT_{rm}}{V_{rm}}(W_{ca,out} - W_{rm,out}) \tag{24}$$

where T_{rm} is the temperature in the return manifold, V_{rm} is the return manifold volume; $W_{rm,out}$ is the return manifold exit air flow rate, which can be calculated by the amended nonlinear nozzle equation modified by the load current [59]:

$$W_{rm,out} = \frac{C_{rm}(\zeta_1 I_{st} + \zeta_2 I_{st}^2 + \zeta_3 I_{st}^3 + \zeta_4 I_{st}^4 + \zeta_5 I_{st}^5 + \zeta_6)(p_{rm} - p_{atm})}{\sqrt{RT_{rm}}} \sqrt{\gamma\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}} \tag{25}$$

where C_{rm} is the flow rate coefficient, T_{rm} is the temperature at the return manifold, ζ_1~ζ_6 are the parameters to be identified.

3. Observer design based on Kalman filter

3.1. Nonlinear dynamic equation for the PEMFC system

A fifth-order nonlinear dynamic model of fuel cell system is obtained by integrating the component models in Eqs. (1)–(25), in which angular velocity of air compressor ω_{cp}, supply manifold pressure p_{sm}, oxygen mass inside the cathode m_{O_2}, nitrogen mass inside the cathode m_{N_2}, and the return manifold pressure p_{rm}, are state variables. The stack current I_{st}, cathode inlet relative humidity $\phi_{ca,in}$, supply manifold temperature T_{sm} and return manifold temperature T_{rm} are considered as input variables. The system dynamic equations are shown as follow:

$$x = [\omega_{cp},\ p_{sm},\ m_{O_2},\ m_{N_2},\ p_{rm}]^T \tag{26.a}$$

$$u = [I_{st},\ \phi_{ca,in},\ T_{sm},\ T_{rm}]^T \tag{26.b}$$

$$\dot{x} = f(x,u) = \begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\\ \dot{x}_4\\ \dot{x}_5\end{bmatrix} = = \begin{bmatrix}f_1(x,u)\\ f_2(x,u)\\ f_3(x,u)\\ f_4(x,u)\\ f_5(x,u)\end{bmatrix} \tag{26.c}$$

$$y = h(x) = [\omega_{cp},\ p_{sm},\ p_{rm}]^T \tag{26.d}$$

where $x \in \Re^5$ is the state vector, $y \in \Re^3$ is the output vector that can be measured by the periphery sensors, $u \in \Re^4$ is the input vector; $f(\cdot)$ contains the nonlinear dynamics of the model presented in Section 2.2 and listed in Appendix A. Because of the hermetic structure of the PEMFC system, it's almost impossible to measure the internal mass of oxygen and nitrogen in the vehicular system. Therefore, it's necessary to develop internal state observer on account of the peripherally measurable signals. In this study, as illustrated in Fig. 1, measurable signals include the compressor rotation speed ω_{cp} (transmitted by compressor controller), supply manifold pressure p_{sm}, return manifold pressure p_{rm}, inlet relative humidity $\phi_{ca,in}$, supply manifold temperature T_{sm}, return manifold temperature T_{rm} and load current I_{st}.

3.2. Internal state estimation of the cathode side

In Kalman filter observers, EKF is the first application in the internal state estimation of fuel cells. Nevertheless, the EKF performs Taylor expansion of the nonlinear model near the predefined points and takes the first-order approximation as the approximate expression to linearize

the model. In that case, additional linearization errors would be introduced, and it's challenging to get the derivation of the Jacobi matrix of higher-order and complex PEMFC systems. The UKF rejects the traditional method with linearizing nonlinear functions and applies the probability density function of the Gaussian density approximation state to dispose of the nonlinear transfer difficulty of the mean and covariance, effectively overcoming low precision and reduced stability of EKF. In contrast to the UKF, the CKF applies the spherical-radial cubature rule to compute multivariate moment integrals encountered in the nonlinear Bayesian filter [61]. Analogously, this cubature rule is derivative-free, and high-dimensional nonlinear filtering problems can also be solved with this approach. Besides, the state estimator based on the CKF with radial-spherical cubature rule is utilized to accomplish the purpose of higher accuracy and better stability than the UKF [62,63]. To highlight the better estimation performance with adaptive CKF, the performance of different observers needs to be compared. Therefore, in this section, the UKF observer for internal state estimation inside cathode is first introduced, and then the adaptive CKF observer is described.

3.2.1. State estimation based on unscented Kalman filter

To employ the UKF approach to design observer, a nonlinear discrete-time dynamical system derived from time integration of continuous system equations that accomplished by a fourth-order Runge-Kutta method with a constant time step is considered:

$$\begin{cases} x_{k+1} = F(x_k, u_k) + \varpi_k \\ y_k = h(x_k) + v_k \end{cases} \tag{27}$$

For the UKF observer design, ϖ_k and v_k are used to represent process and measurement noises at the sampling time k. According to the classic Kalman filter assumptions [23], ϖ_k and v_k are assumed as zero-mean Gaussian white noises with covariances of Q_x and R_x, respectively.

The steps of the UKF algorithm is illustrated as follow:

Step 1: Initialization for the states and their covariances:

$$\hat{x}_{0|0} = E(x_0) \tag{28.a}$$

$$P_{0|0} = E[(x_0 - \hat{x}_0)(x_0 - \hat{x}_0)^T] \tag{28.b}$$

Step 2: The unscented transform is applied to transform n dimensional state vector to $2n + 1$ dimensional state vector:

$$x^i_{k-1|k-1} = \begin{cases} \hat{x}_{k-1/k-1}, & i = 0 \\ \hat{x}_{k-1/k-1} + \sqrt{(n+\lambda)}\, S^i_{k-1|k-1}, & i = 1...n \\ \hat{x}_{k-1/k-1} - \sqrt{(n+\lambda)}\, S^i_{k-1|k-1}, & i = n+1...2n \end{cases} \tag{29}$$

where S^i_{k-1} stands for the i-column of Cholesky decomposition matrix of state variables covariance $\hat{P}_{x,k-1|k-1}$ at the k time point. $\lambda = \alpha^2(n + \kappa) - n$ is a scaling parameter, α decides the spread of the sigma points and is 0.01 in this paper; κ is degree of freedom of sigma points and is set to 0.

Step 3: According to the discrete-time system equation and the process noise covariance, the predicted state $x^i_{k|k-1}$ and priori error covariance $\hat{p}_{x,k|k-1}$ firstly update:

$$x^i_{k|k-1} = F(x^i_{k-1/k-1}, u_{k-1}) \tag{30.a}$$

$$\hat{x}_{k|k-1} = \sum_{i=0}^{2n} w^i_m x^i_{k|k-1} \tag{30.b}$$

$$\hat{p}_{x,k|k-1} = \sum_{i=0}^{2n} w^i_c (x^i_{k|k-1} - \hat{x}_{k|k-1})(x^i_{k|k-1} - \hat{x}_{k|k-1})^T + Q_x \tag{30.c}$$

The weighting parameters can be obtained by Eq (30.d):

$$\begin{cases} w^0_m = \frac{\lambda}{n+\lambda} \\ w^0_c = \frac{\lambda}{n+\lambda} + (1 - \alpha^2 + \beta) \\ w^i_m = w^i_c = \frac{1}{2(n+\lambda)}, \; i = 1...2n \end{cases} \tag{30.d}$$

β is the prior information and takes 2 in the Gaussian distribution.

Step 4: Generate new sigma points at time instant $k|k-1$ with the determination of updated state variables covariance $\hat{p}_{x,k|k-1}$:

$$x^i_{k|k} = \begin{cases} \hat{x}_{k/k-1}, & i = 0 \\ \hat{x}_{k/k-1} + \sqrt{(n+\lambda)}\, S^i_{k|k-1}, & i = 1...n \\ \hat{x}_{k/k-1} - \sqrt{(n+\lambda)}\, S^i_{k|k-1}, & i = n+1...2n \end{cases} \tag{31}$$

Step 5: Measurement update based on the output equation and the measured noise:

$$y^i_{k|k-1} = h(x^i_{k|k}) \tag{32.a}$$

$$\hat{y}_{k|k-1} = \sum_{i=0}^{2n} w^i_m y^i_{k|k-1} \tag{32.b}$$

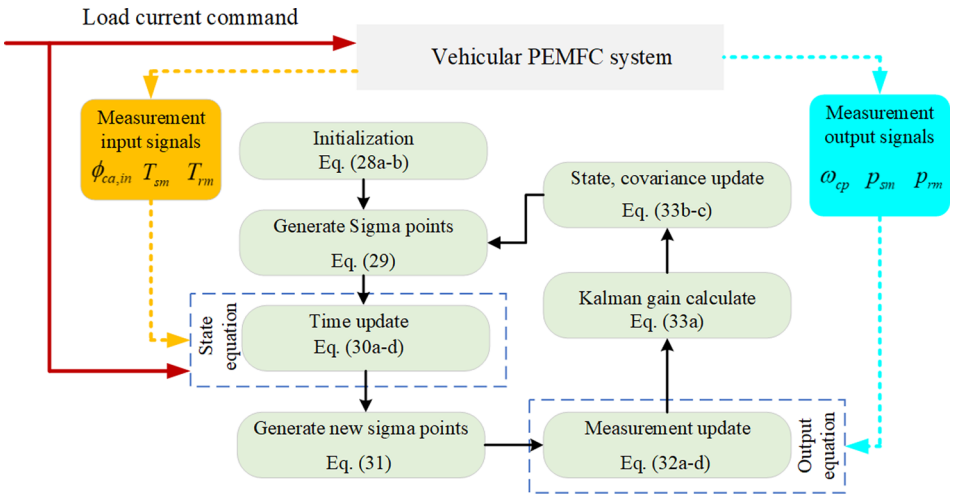


Fig. 2. Schematic diagram of fuel cell internal states estimation based on the UKF algorithm.

$$\hat{P}_{yy,k|k-1} = \sum_{i=0}^{2n} w_c^i (y_{k|k-1}^i - \hat{y}_{k|k-1})(y_{k|k-1}^i - \hat{y}_{k|k-1})^T + R_x \tag{32.c}$$

$$\hat{P}_{xy,k|k-1} = \sum_{i=0}^{2n} w_c^i (x_{k|k-1}^i - \hat{x}_{k|k-1})(y_{k|k-1}^i - \hat{y}_{k|k-1})^T \tag{32.d}$$

Step 6: Calculating Kalman gain updates the system states and error covariance according to the system measurement:

$$K = \hat{P}_{xy,k|k-1}\hat{P}_{yy,k|k-1}^{-1} \tag{33.a}$$

$$\hat{x}_{k|k} = x_{k|k}^i + K(y_k - \hat{y}_{k|k-1}) \tag{33.b}$$

$$\hat{P}_{x,k|k} = \hat{P}_{x,k|k-1} - K\hat{P}_{yy,k|k-1}K^T \tag{33.c}$$

The basic idea used for internal state estimation within cathode is shown in Fig. 2. After initialization, repeating the above Eqs. (29)–(33) could recursively estimate the internal gas mass. However, it should be emphasized that the set of process noise and measurement noise variance determines the accuracy, which generally need to pass through many times after debugging an appropriate value. The basic premise for noise variance debugging is to grasp the effect of these two variance matrices on the Kalman gain matrix so that the process noise covariance in the state equation should not be too big. Otherwise, the fluctuation in parameter estimation is very severe and even challenging to converge.

3.2.2. State estimation based on adaptive cubature Kalman filter

In this paper, the three-degree spherical-radial cubature rule is employed to calculate the mean and covariance of state variables. The cubature point and weighting coefficients are illustrated as follow:

$$\xi^{(i)} = \sqrt{\frac{1}{m}}\,[1]_i,\ \mu_i = \frac{1}{m},\ i = 1, 2, \ldots m,\ m = 2n \tag{34}$$

where $\xi^{(i)}$ is the cubature points, μ_i is the weighting coefficients, and $[1]_i$ stands for ith the column of following matrix:

$$[1]_1 = [diag(1, \ldots 1), diag(-1,\ldots -1)]_i \tag{35}$$

The noise covariance in the UKF observer is assumed to be constant, while the practical ambient noise of the fuel cell system cannot be represented with such condition. Besides, the convergence of the Kalman filter algorithm is strictly related to the selection of noise variance. It takes a lot of time to obtain the appropriate noise covariances based on the empirical analysis or manual adjustment. Moreover, this method may diverge if the noise covariances are not appropriately chosen. Consequently, it is necessary to adapt the time-varying noise to improve the estimation accuracy and stability of the algorithm further. However, traditional CKF is similar to the UKF that the error covariance is required to be strictly positive to satisfy Cholesky decomposition. Adding adaptive processes could break the positive condition of the covariance matrix after several times of iteration due to inescapable errors introduced by the rounding and discretization. In this paper, singular value decomposition is proposed to calculate the covariance rather than Cholesky decomposition to increase the stability of the algorithm. With such an improvement, the adaptive process of noise variance is easy to implement.

Assume that $A \in R^{n\times n}$, the singular value decomposition of matrix A can be described as:

$$A = USV^T \tag{36}$$

where $U \in R^{n\times n}$, $V \in R^{n\times n}$, $S = diag(s_1, s_2 \ldots s_n)$ are singular values.

The algorithm of SVD-ACKF is then summarized as follow:

Step 1: Initialization:

$$\hat{x}_{0|0} = E(x_0) \tag{37.a}$$

$$P_{0|0} = E[(x_0 - \hat{x}_0)(x_0 - \hat{x}_0)^T] \tag{37.b}$$

Step 2: Cubature points generate, state and covariance update:

$$\hat{P}_{x,k-1|k-1} = U_{k-1|k-1}S_{k-1|k-1}V_{k-1|k-1}^T \tag{38.a}$$

$$x_{k-1|k-1}^i = \hat{x}_{k-1|k-1} + U_{k-1|k-1}\sqrt{S_{k-1|k-1}}\,\xi^{(i)} \tag{38.b}$$

$$x_{k|k-1}^i = F(x_{k-1/k-1}^i, u_{k-1}) \tag{38.c}$$

$$\hat{x}_{k|k-1} = \sum_{i=1}^{m} \mu_i x_{k|k-1}^i \tag{38.d}$$

$$\hat{p}_{x,k|k-1} = \sum_{i=1}^{m} \mu_i (x_{k|k-1}^i - \hat{x}_{k|k-1})(x_{k|k-1}^i - \hat{x}_{k|k-1})^T + Q_{x,k-1} \tag{38.e}$$

Step 3: Select new cubature points and measurement update:

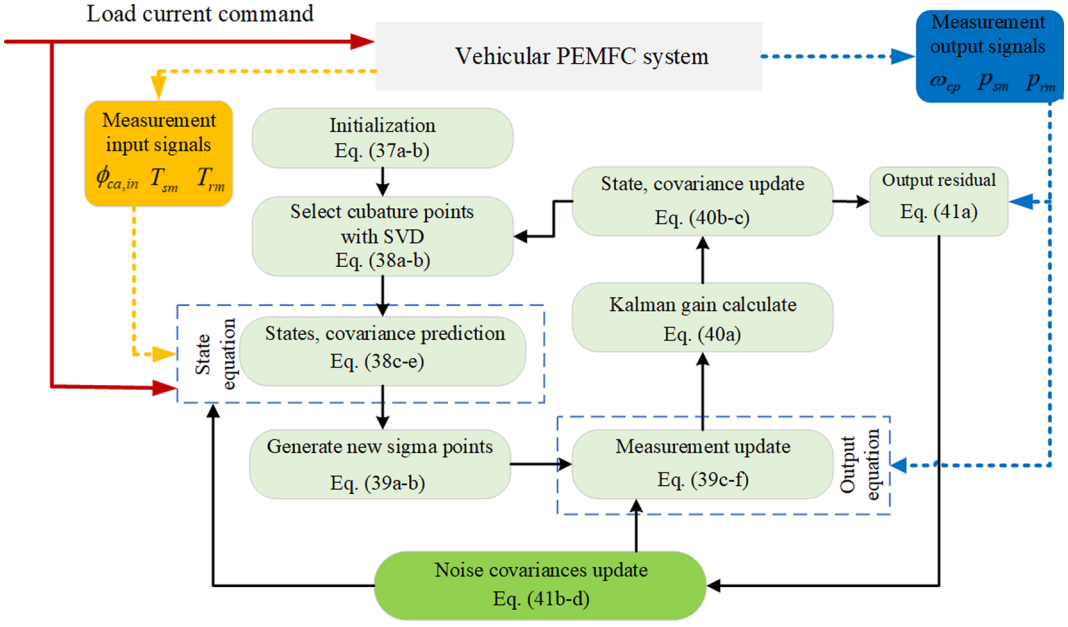


Fig. 3. Schematic diagram of the proposed SVD-ACKF-based internal states estimation algorithm.

$$\hat{p}_{x,k|k-1} = U_{k|k-1} S_{k|k-1} V_{k|k-1}^T \tag{39.a}$$

$$x_{k|k}^i = \hat{x}_{k|k-1} + U_{k|k-1}\sqrt{S_{k|k-1}}\,\xi^{(i)} \tag{39.b}$$

$$y_{k|k-1}^i = h(x_{k|k}^i) \tag{39.c}$$

$$\hat{y}_{k|k-1} = \sum_{i=1}^{m} \mu_i y_{k|k-1}^i \tag{39.d}$$

$$\hat{P}_{yy,k|k-1} = \sum_{i=1}^{m} \mu_i (y_{k|k-1}^i - \hat{y}_{k|k-1})(y_{k|k-1}^i - \hat{y}_{k|k-1})^T + R_{x,k-1} \tag{39.e}$$

$$\hat{P}_{xy,k|k-1} = \sum_{i=1}^{m} \mu_i (x_{k|k-1}^i - \hat{x}_{k|k-1})(y_{k|k-1}^i - \hat{y}_{k|k-1})^T \tag{39.f}$$

Step 4: Gain, states and covariance update:

$$K = \hat{P}_{xy,k|k-1} \hat{P}_{yy,k|k-1}^{-1} \tag{40.a}$$

$$\hat{x}_{k|k} = x_{k|k}^i + K(y_k - \hat{y}_{k|k-1}) \tag{40.b}$$

$$\hat{P}_{x,k|k} = \hat{P}_{x,k|k-1} - K\hat{P}_{yy,k|k-1} K^T \tag{40.c}$$

Step 5: Noise covariances update by learning history data:

$$e_k = y_k - h(\hat{x}_{k|k}) \tag{41.a}$$

$$C_k = \frac{1}{L} \sum_{k-L+1}^{k} e_k e_k^T \tag{41.b}$$

$$Q_k = K C_k K^T \tag{41.c}$$

$$R_k = C_k + H P_{k|k-1} H^T \tag{41.d}$$

where e_k is the output residual of the fuel cell system model at time step k, and C_k is an approximation to the covariance of output residual at time step k; L is 20, $H = [1, 0, 0, 0, 0; 0, 1, 0, 0, 0; 0, 0, 0, 0, 1]$.

The schematic diagram of the proposed SVD-ACKF-based internal state estimation algorithm is shown in Fig. 3. After initialization, the estimated state variables and error covariance can be calculated by the prediction and update process. The output residual of measurable signals is calculated based on the fuel cell system. Afterward, the process noise covariance and measurement noise covariance are adaptive updated and then utilized for the subsequent prediction and update process. The mass of oxygen and nitrogen inside the cathode can be estimated by repeating the above process.

4. Experiment and results

4.1. Fuel cell system test bench

As shown in Fig. 4, the fuel cell system test bench is established to obtain the proposed parameter identification. The test bench consists of hydrogen supply assembly, an electronic load, a high voltage power supply, and a LabView-based virtual host computer. The hydrogen supply assembly provides adequate hydrogen for the fuel cell system and monitors the surrounding environment hydrogen concentration in real-time. An alarm would appear when the hydrogen concentration exceeds the limited value, and the hydrogen supply is cut off immediately at the same time. The electronic load is responsible for loading the fuel cell system and adjust the current according to the instructions sent from the host computer. During the development of a fuel cell system, the air compressor and heat unit are powered by high-voltage supply instead of the fuel cell stack, while the HCP, fans, and water pumps are powered by low-voltage supply. The fuel cell system controller ensures the normal operation of the fuel cell and records signals from critical sensors. Restricted by the channels of the fuel cell system controller, excess data that cannot be recorded is obtained by a data acquisition unit (DAU) in the National Instrument Chassis. The measured data in the controller and DAU are transmitted to the computer through the controller area network (CAN) bus driven by the LabView program and CAN communication unit. Moreover, the required load profile can be sent to the controller through the host computer. Given the measured data, the internal states can be obtained by the estimated algorithm in the controller and then transmitted through CAN bus to the computer for real-time display.

4.2. Parameters identification and results

It's imperative to obtain critical parameters of the fuel cell system

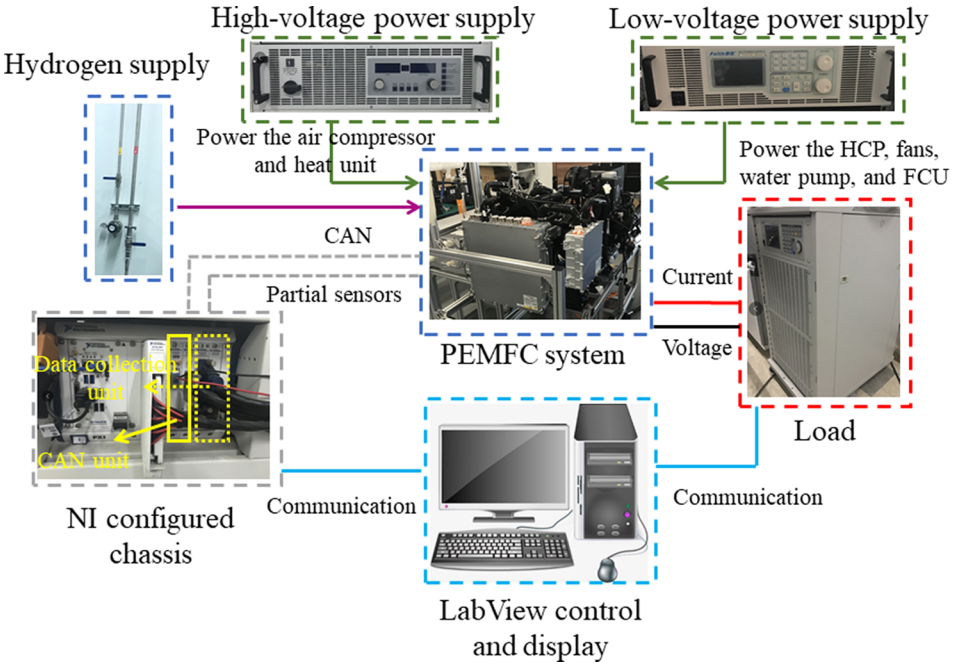


Fig. 4. PEMFC system test bench and data acquisition module in this paper.

model using periphery signals such as inlet pressure, current, and temperature. In this paper, a broader dynamic current profile that emulates the vehicular working range is carried to excite the fuel cell system to get the data used for identification, and the stack current and voltage are shown in Fig. 5. Given the sampled information, the model parameters to be identified can be capsulated into a vector $\theta \in \Re^{13}$:

$$\theta = [c_1, c_2, c_3, c_4, c_5, \zeta_1, \zeta_2, \zeta_3, \zeta_4, \zeta_5, \zeta_6, k_{sm,out}, k_{ca,out}]^T \tag{42}$$

During the identification process, the objective is to find the model suitable parameters so as to minimize the root-mean-squared error between the model output vector and measured values. For the given working profile, such identification problem can be mathematically interpreted as the following equation:

$$\hat{y} = h(x_k, \theta_{opt}) \tag{43}$$

$$\min fit(\theta_{opt}) = \sum_{k=1}^{M} [\hat{y} - y_k]^2 \tag{44}$$

where $fit(\cdot)$ is the adaptation function and M is the number of the sampling points, θ_{opt} represents the optimal identified parameters, y_k is the measured output vector. This is a nonlinear least square problem that the model parameters can be solved using the Matlab parameter identification toolbox. The physical parameters of gas and the fuel cell system are shown in Table 1. Since the air compressor model involves parameters such as the rotational inertia and motor constants, which are generally not provided by the compressor manufacturer and particularly hard to measure. Therefore, the physical parameters of the compressor are modified appropriately based on Refs. [14,22,55]. Although there are inescapable errors with the actual parameters, these errors will eventually be equivalent to the parameters to be identified. And the characteristic function of the air compressor out mass flow can be optimized based on the tested Map data in advance. The physical parameters of the stack are calculated according to the type, width, and depth of the gas channel provided by the manufacturer's manual. Moreover, the manifold volumes are obtained by cross-sectional area and length. Based on the above experimental and optimization specifications, the identified parameters are listed in Table 2.

Fig. 6 compares the measured values and model output values and exhibits the relative error of each variable, which indicates that the simulated curves fit the experimental data well. The maximum and mean relative errors of compressor angular velocity are 5.48% and 1.5041%, respectively. Besides, the maximum relative errors of supply manifold pressure and return manifold pressure are within 5%. The modeling errors become relatively large when the load variation is enormous due to the hysteresis effect of compressor caused by the open-loop control algorithm based on the polynomial form. Compared with the closed-loop control algorithm, it has poor dynamic response capability. While the discussion of the control algorithm of the compressor is not contained in this paper, which will be discussed in a subsequent article. On the other hand, the original compressor MAP, including measured flow rate, angular velocity, and outlet pressure, are also fitted by a polynomial to facilitate matrix calculation in the observer, which will introduce additional errors to the model. But overall, it is considered that this system model could accurately follow the measurement trajectories in this working profile.

Fig. 7(a) illustrates the changes in the air excess ratio calculated by the measured intake air mass flow and the air mass flow required for complete reaction under specific current. The air excess ratio increases suddenly in the 580 s and 1580 s because of the response time difference between the load current and the air compressor, resulting in more air intake than consumption. Fig. 7(b) shows the dynamic simulation results of the mass of oxygen and nitrogen inside the cathode under this working cycle. The nitrogen mass fluctuates as same as manifold pressure and current due to no chemical depletion, while the oxygen mass firstly increases with increasing the load current, and then decreases slightly due to the drop of the air excess ratio. Besides, the oxygen mass improves suddenly with a considerable current load variation at 1580 s due to the changes in air excess ratio and larger current before load reduction.

In this paper, the relative humidity inside the cathode is supposed to be constant, as same as the papers [13,14,53,56,57]. However, these references assumed that the relative humidity inside the cathode equals 1, which is not suitable in the practical fuel cell system because the stack is prone to flooding under such conditions. In the above working profile, the relative inlet humidity is mostly within the range [40%, 60%], and inlet gas was humidified by the exhausted gas. Hence the relative humidity inside the cathode is assumed to be 0.8 in this paper. Besides, through the simulation experiment, it is found that the humidity has a small influence on internal states (10% relative variation of relative humidity leads to 2% variation on oxygen mass, and nitrogen is even lower). Therefore, it is assumed that the performance of the fuel cell is relatively ideal, and the actual humidity inside the cathode could reach the expected humidity.

To the best of our knowledge, due to the great difficulty of measuring the internal states within the fuel cell, offline values obtained based on models are overwhelmingly applied for observer design and validation. Therefore, as in other papers, the mass of oxygen and nitrogen in the model, as shown in Fig. 7(b), is considered as "real" value. To further explain the rationality of the dynamic characteristics inside the cathode, the fuel cell voltage was employed for verification. The voltage model is established according to Ref. [64] and calculated as follow:

$$\begin{cases} V_{stack} = N \cdot V_{cell} = N \cdot (V_{nt} - V_{act} - V_{ohm} - V_{conc}) \\ V_{nt} = 1.229 - (8.5 \times 10^{-4}) \times (T_{st} - 298.15) + \frac{RT_{st}}{2F} \times \ln(p_{H_2} \times \sqrt{p_{O_2}}) \\ V_{act} = \theta_1 + \theta_2 T_{st} + \theta_3 T_{st} \ln I_{st} + \theta_4 T_{st} \ln C_{O_2} \\ V_{ohm} = (\theta_5 + \theta_6 T_{st} + \theta_7 I_{st}) \cdot I_{st} \\ V_{conc} = \theta_8 \cdot \exp(\theta_9 I_{st}) \\ C_{O_2} = 1.97 \times 10^{-7} \times p_{O_2} \times \exp(498/T_{st}) \end{cases} \tag{45}$$

where V_{stack} is fuel cell stack voltage, V_{cell} is the average single cell voltage, V_{nt} means the Nernst voltage, V_{act} stands for the activation voltage loss, V_{ohm} is the ohmic voltage loss, and V_{conc} is the concentration voltage loss. p_{O_2} is oxygen partial pressure inside the fuel cell and can be calculated by the Eq. (20). p_{H_2} is the hydrogen pressure inside the fuel cell, which is approximated as the average pressure of anode inlet and outlet. Data from 100 s to 750 s were selected to identify voltage model parameters with a hybrid genetic algorithm/particle swarm optimization method to reduce risks of being trapped into local optima [65]. The identified electrochemical parameters of the voltage model are listed in Table 3. As can be seen from Fig. 8, the maximum error of

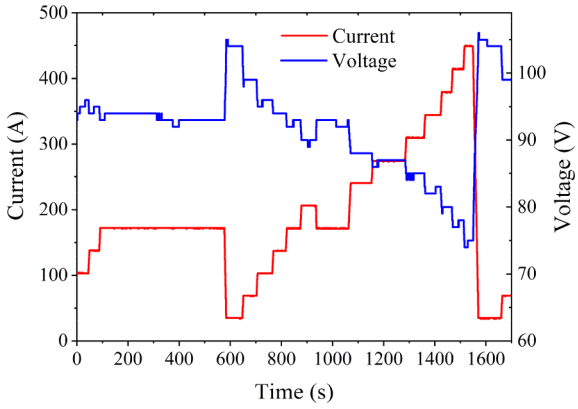


Fig. 5. Current and voltage profiles for fuel cell system to acquire identification data.

Table 1
Physical parameters of the fuel cell system.

Component	Symbol	parameter	Value
Air compressor	R_{cm}	Motor constant	1.25×10^{-4} ohm
	K_t	Motor constant	0.0059 Nm/A
	K_v	Motor constant	0.0059 V/(rad/s)
	η_{cm}	Compressor motor mechanical efficiency	0.98
	η_{cp}	Compressor efficiency	0.8
	J_{cp}	Compressor rotational inertia	0.03 g.m^2
	γ	Air specific heat ratio	1.4
	C_p	Air specific heat capacity	1004 J/(kg.K)
	B_{00}	Air compressor parameters	0.028262
	B_{10}	Air compressor parameters	3.9442×10^{-5}
	B_{20}	Air compressor parameters	8.956×10^{-8}
	B_{01}	Air compressor parameters	-4.3539×10^{-7}
	B_{11}	Air compressor parameters	-7.5885×10^{-10}
	B_{02}	Air compressor parameters	2.8777×10^{-12}
Manifold	V_{sm}	Supply manifold volume	0.0057 m^3
	V_{rm}	Return manifold volume	0.0043 m^3
	C_{rm}	Flow rate coefficient in return manifold	0.01488
Cathode volume	T_{st}	Stack temperature	348 K
	V_{ca}	Stack cathode volume	0.00548 m^3
	N	Serial connected cell number	130
	A_{cell}	Active area of each cell	340 cm^2
Gas parameter	p_{atm}	Air pressure	1.013×10^5 pa
	T_{atm}	Air temperature	298 K
	$Y_{O_2,ca,in}$	oxygen molar ratio at cathode inlet	0.21
	R_a	Air gas constant	286.9 J/(mol.K)
	R_{O_2}	Oxygen gas constant	259.8 J/(mol.K)
	R_{N_2}	Nitrogen gas constant	296.8 J/(mol.K)
	R_v	Vapor gas constant	461.5 J/(mol.K)
	M_{air}	Air molar mass	28.84 g/mol
	M_{O_2}	Oxygen molar mass	32 g/mol
	M_{N_2}	Nitrogen molar mass	28 g/mol
	M_v	Vapor molar mass	18 g/mol
	F	Faraday constant	96,487 C/mol

the terminal voltage is 6.25% and appears in the initial stage of operation. In the stable condition, the average relative error of the output voltage is 0.61%, and the overall relative error is within 4%, indicating that the established fuel cell system model and parameters are reasonable and effective.

4.3. Observers validation

4.3.1. Performance comparison of the observers

In this section, the effectiveness of the proposed algorithm is examined over the above working profile. In order to fully demonstrate the superiority of the adaptive process, the performance of the SVD-ACKF observer is compared with the UKF observer and SVD-CKF observer with no adaptive process. The current, manifold pressure, relative inlet humidity, and angular speed of compressor are measured by the sensors including the noise. The initial values of the model and observer are supposed to be the same, as noted in Table 4. Since the actual initial states are challenging to acquire, $P_{0|0}$ is a given matrix in this paper. For the UKF and SVD-CKF observer, Q_x and R_x are assumed to maintain constant during the estimation process and are considered to be the initial value of the error covariance iteration of the SVD-ACKF observer. These parameters are selected as follow:

$$P_{0|0} = \begin{bmatrix} 1e^{-9}, 0, 0, 0, 0 \\ 0, 1e^{-9}, 0, 0, 0 \\ 0, 0, 1e^{-9}, 0, 0 \\ 0, 0, 0, 1e^{-8}, 0 \\ 0, 0, 0, 0, 1e^{-9} \end{bmatrix}, \ Q_x = \begin{bmatrix} 1e^{-5}, 0, 0, 0, 0 \\ 0, 1e^{-3}, 0, 0, 0 \\ 0, 0, 3e^{-9}, 0, 0 \\ 0, 0, 0, 1.2e^{-8}, 0 \\ 0, 0, 0, 0, 1e^{-6} \end{bmatrix}, \ R_x = \begin{bmatrix} 30,0,0 \\ 0,30,0 \\ 0,0,0.01 \end{bmatrix} \tag{46}$$

To characterize the estimation accuracy, the root-mean squared error (RMSE) and relative percentage error with the definition of

$$error = \left| \frac{m_{i,k} - \hat{m}_{i,k}}{m_{i,\max}} \right| \times 100 \tag{47}$$

are used, where $m_{i,\max}$ is the maximum of value of $m_{O_2,k}$ and $m_{N_2,k}$ for all possible k.

Fig. 9 compared the estimated values of different observers and their "real" values obtained from the model, in which the black line represents the theoretical value, and blue line, orange line, and red line indicate the estimation with UKF, SVD-CKF, and SVD-ACKF, respectively. It's apparent that all the observers are capable of predicting the "real" oxygen mass in a quick way. Moreover, the estimation errors of oxygen mass with the UKF are maximum compared with other two observers. Meanwhile, the steady estimations of SVD-CKF and SVD-ACKF converge to a small neighborhood of their "true" values, as evidenced by the steady relative error of less than 3% and 0.2%. Besides, perfect estimation with close to zero steady-state error is possible for nitrogen estimation cooperated with both SVD-CKF and SVD-ACKF. In summary, the estimated results of the three observers are recorded in Table 5. These results indicate that the CKF observer presents better estimation accuracy and robustness against noises. Besides, adaptive adjusting covariance of the process and measurement are profoundly beneficial to the improvement of the internal state estimation accuracy.

4.3.2. Robustness against uncertainties in initial values

Among the initial values of the system state variables, the compressor angular velocity, inlet manifold pressure, and exhaust manifold pressure can be measured by the sensors and therefore are the same as those in the model. For the other two states, the initial values may have a relatively large error because they are unmeasurable. To evaluate the potential negative influence generated by the initial errors, six levels of initialization errors of ± 10%, ± 30%, ± 50% are utilized for the oxygen and nitrogen mass estimations with the SVD-ACKF due to its perfect estimation accuracy. The estimation results of the states inside the cathode are presented in Fig. 10. As can be seen from the figure, although the relative errors of the mass estimation of oxygen and nitrogen are relatively large at the beginning of this working profile, the estimated oxygen mass converges to the "real" value within 10 s. In comparison, the estimated nitrogen mass converges to the "truth" value within 3 s. Therefore, the SVD-ACKF algorithm can quickly compensate for the initial errors and track to reference values rapidly and nearly throughout the whole process.

Table 2
Identified fuel cell system model parameters.

c_1	c_2	c_3	c_4	c_5	
0.0326	-5.6113×10^{-3}	1.458×10^{-8}	4.2199×10^{-11}	1.1604	
ζ_1	ζ_2	ζ_3	ζ_4	ζ_5	ζ_6
-2.3198×10^{-3}	1.0211×10^{-6}	1.7490×10^{-9}	5.0788×10^{-13}	1.0287×10^{-15}	0.0228
$k_{ca,out}$			$k_{sm,out}$		
3.8846×10^{-6}			6.2194×10^{-6}		

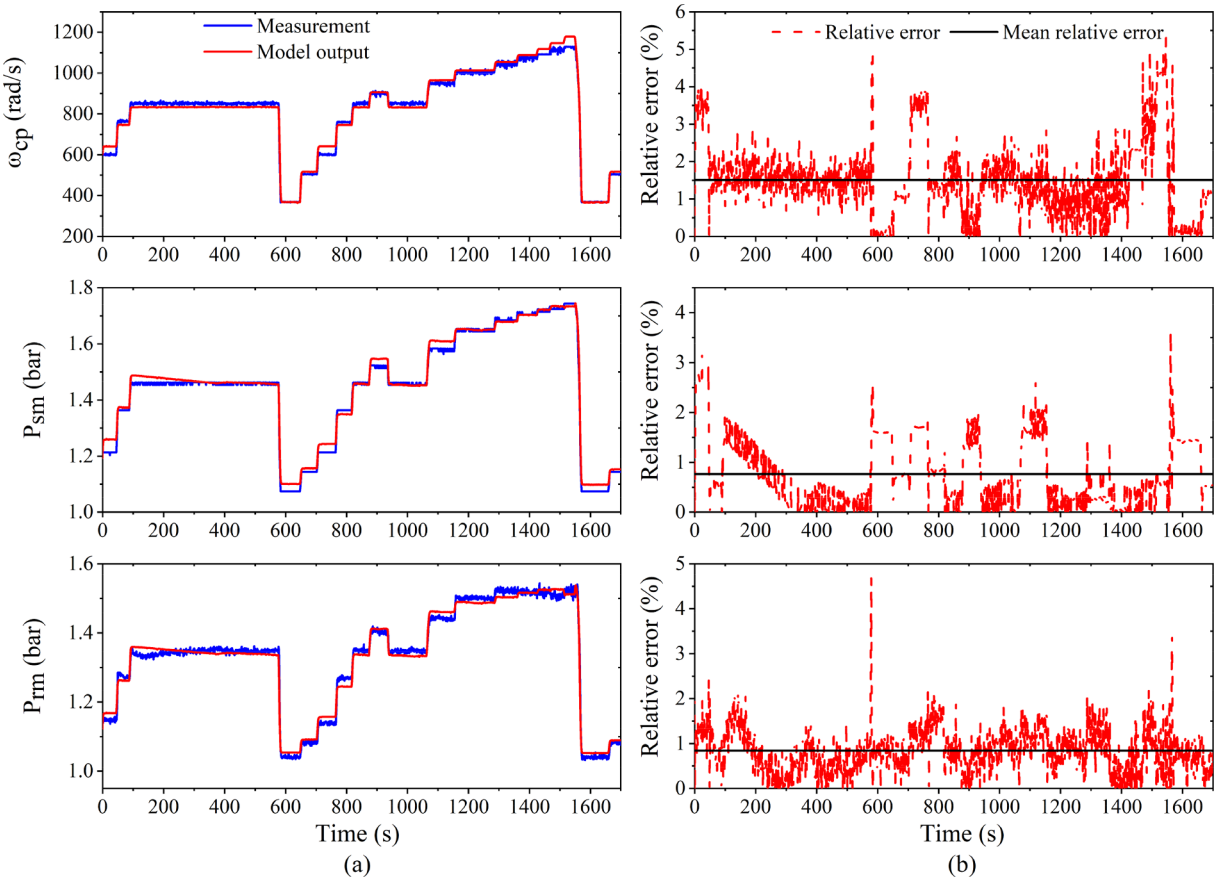


Fig. 6. Results of model identification; (a) Measured and simulated values in this test profile; (b) Relative error of model output.

4.3.3. Robustness against uncertainties in model parameters

As the working time of the fuel cell is prolonged, various parameters in the system will change correspondingly, so there are uncertainties of model parameters in the actual fuel cell system. To detect the robustness of the observers against the uncertainties of system parameters, the volumes and orifice coefficients of the supply manifold and return manifold, as well as stack temperature that determine the performance of the observers, are employed, respectively. In this case, the nominal values and acceptable variations of these system parameters are displayed in Table 6. Besides, the initial values of the state variables in Table 4 are used. The obtained estimation results are given in Fig. 11. Despite variations on the system parameters, the estimation for the mass of oxygen and nitrogen can still follow their "real" values under the first fourth simulations. In detail, even though with the 10% variations on the gas manifold and 5% variations on the orifice coefficients of the manifolds, the estimation errors of oxygen mass with SVD-ACKF are reduced to around 0.5% within 10 s and then gradually converges towards the origin with zero errors. With some fluctuations at the beginning of the algorithm running, the estimations of nitrogen mass exhibit similar performance in the convergence and precision. In

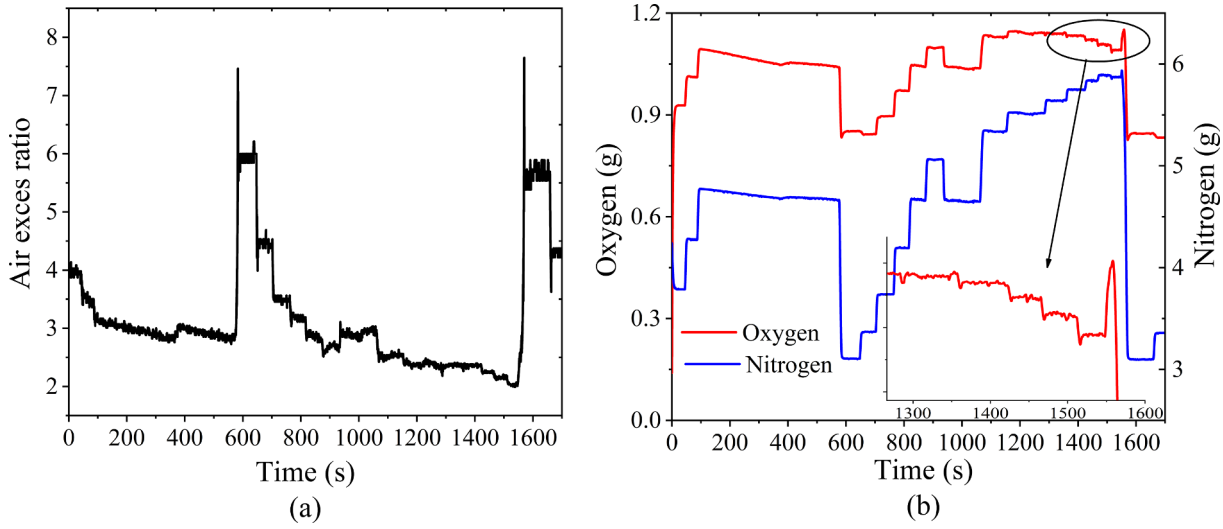


Fig. 7. (a) The changes of the air excess ratio; (b) The dynamic process inside the cathode.

Table 3
Identified electrochemical parameters.

θ_1	θ_2	θ_3	θ_4	θ_5
0.4883	-5.9237×10^{-4}	1.7546×10^{-4}	-8.7752×10^{-5}	0.4753
θ_6	θ_7		θ_8	θ_9
-1.3655×10^{-3}	6.8741×10^{-7}		1.1341×10^{-5}	3.4749×10^{-3}

general, compared with the other two observers, the estimation with SVD-ACKF shows the best robustness against the uncertainties of model parameters.

Besides, as can be seen from the last two subfigures, all observers are sensitive to the disturbance of the fuel cell temperature uncertainty. Accurately, the SVD-ACKF observer could track trends in the mass of oxygen and nitrogen within steady-state error less than 5% under 5% variation on stack temperature. Nevertheless, all the observers show a low accuracy when the relative error of the stack temperature is 10%. At this point, it is necessary to control the relative error of the stack temperature in a reasonable range to avoid unexpected results.

4.3.4. Robustness against different power requirements

In order to further verify the robustness of the observers against different power requirements, the observer experiments were carried out under a test protocol of Tongji University that proposed based on the New European Drive Cycle (NEDC). The NEDC is mainly applied for assessing the emissions and fuel economy of passenger vehicles driving in European cities, as shown in Fig. 12a, which incorporates four repeated urban driving cycles and one suburb driving cycle, including startup-shutdown, idling running, partial power running, full power running, and dynamic loading running, the whole duration of the protocol is about 1200 s. It is considering that when the vehicle is under idling running, the stack needs to supply power to auxiliary components such as air compressor and hydrogen cycle pump. Also, the stack should be avoided in open-circuit voltage to prevent irreversible damage caused by over-potential. Therefore, the idle speed running in Fig. 12a is replaced by the power of 2.5 kW in Fig. 12b and relative current density are also illustrated. Fig. 12c and d, refer to the mass estimation of oxygen and corresponding relative error. On the other hand, Fig. 12e and f refer to the nitrogen estimation. The results indicate that the fast convergence of all observers and their high estimation accuracy are both evident, in which the proposed SVD-ACKF shows the best performance in face of different typical power requirements.

Table 4
Initial values of internal states in the system model and observers.

	Parameter	Value
Initial values in the observer and model	ω_{cp}	602.46133 rad/s
	P_{sm}	121,325 Pa
	m_{O_2}	0.14 g
	m_{N_2}	4.2 g
	P_{rm}	115,405 Pa

5. Discussion

From the results mentioned above, the model-based SVD-ACKF observer could effectively solve the problem that the internal gas mass inside the cathode is unmeasurable. However, the practical application of the observer proposed in this paper or other studies is still limited.

The performance of the observer is highly dependent on the fuel cell model selected. But the fuel cell internal states involve multiple coupling characteristics and different time scales. For example, the temperature usually displays lower dynamic response compared to others such as pressure and gas flow, and membrane water content is directly related to the output voltage rather than other measured variables. Thus, the current model applied by the observer is usually *iso*-thermal, single-phase, very few works consider the temperature, water phase change, along with variables that are associated with aging and irreversible recession in the PEMFC. Indeed, hydrothermal management in

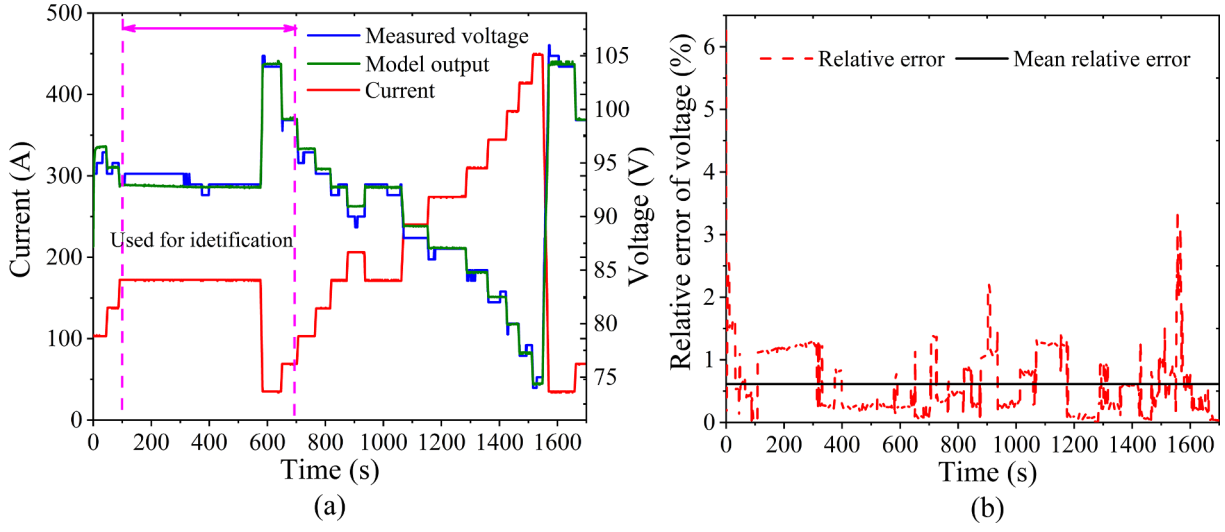


Fig. 8. (a) Current and voltage profiles for voltage model identification and validation; (b) The relative error of the fuel cell system output voltage.

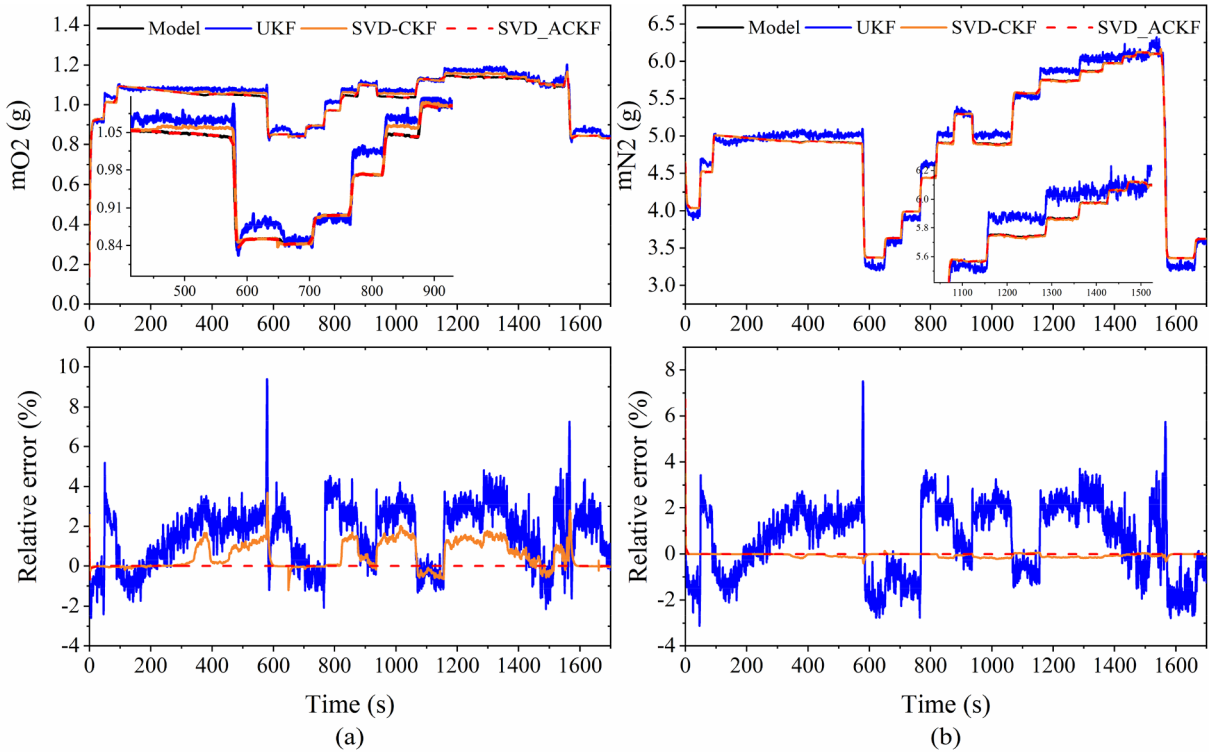


Fig.9. Estimation results of oxygen and nitrogen; (a) estimated oxygen mass and relative error; (b) estimated nitrogen mass and relative error.

Table 5
Comparison for the estimation results.

	Methods	UKF	SVD-CKF	SVD-ACKF
m_{O_2}	Max error	9.39%	3.70%	1.98%
	RMSE	6.27×10^{-10}	9.29×10^{-11}	1.53×10^{-13}
m_{N_2}	Max error	7.51%	6.07%	6.77%
	RMSE	1.09×10^{-8}	5.15×10^{-11}	2.55×10^{-11}

the fuel cell has an essential impact on the performance and lifespan of the whole system. An oversimplified model not only limits the number of observable states but also weakens the observability of the system.

Apart from the limitation of the multi-states coupling, most nonlinear observers are developed with lumped parameter models. In this way, pressure, temperature, and concentration within the fuel cell are considered as uniformly distributed that the states obtained by the observer are the mean values. However, the reactant concentration in the downstream region of the gas channel will be significantly lower than the level in the upstream region due to the existence of the mass transfer process and electrochemical reaction, and the reactant concentration involved in the electrochemical reaction at the catalyst layer is also obviously inferior to that in the gas channel. As a result, local gas starvation and falls in current density caused by poor mass transfer cannot be distinguished.

At present, most studies have adopted nonlinear algorithms (e.g., the UKF and HOSM) to estimate internal states. Among them, under the condition of observability, the HOSM observer has attracted considerable attention because it guarantees fast convergence and relatively high accuracy as long as the sliding mode control parameters are reasonably selected to meet the finite-time convergence requirements. Whereas the UKF observer is sensitive to noise, and unreasonable noise variance setting will lead to poor estimation accuracy. In this paper, the proposed SVD-ACKF observer also has high precision and good robustness against perturbed conditions and similar to the results in other studies related to sliding mode observers [13,14,53–55]. Therefore, based on the same set of data, different observers need to be compared from the perspectives of precision, calculation time, and convergence speed before practical application to select the most suitable algorithm. Furthermore, the sensor signals are influenced by the vehicle environment and electromagnetic compatibility profoundly. Considering that the input signals may have a significant drift, the observer robustness against input drift should be detected subsequently.

The following design of the internal state observer of fuel cell also needs to consider the actual existing conditions of the real vehicle system. For example, the observer proposed in this paper requires a humidity measurement from the cathode inlet, while some practical vehicle fuel cell systems do not possess this sensor for cost savings, observers in some studies that overlook the inlet humidity signal assume that the inlet humidity is 100%, which is inconsistent with the reality. For another example, the distributed state observer in Ref. [20,21] requires to know the molar mass flow of each gas entering the anode and cathode. In contrast, the actual vehicle mass flow sensor only measures the mixed air mass flow rather than the specific gas content, and the hydrogen side is even not equipped with the flow sensor. Of course, with the continuous progress of sensor technology, diverse measurable peripheral signals can significantly improve the observability and estimation accuracy of internal states.

The application limitation of the fuel cell controller should also be considered. For instance, the observer simulations in the Ref. [53] were performed in MATLAB/Simulink environment with the sampling time

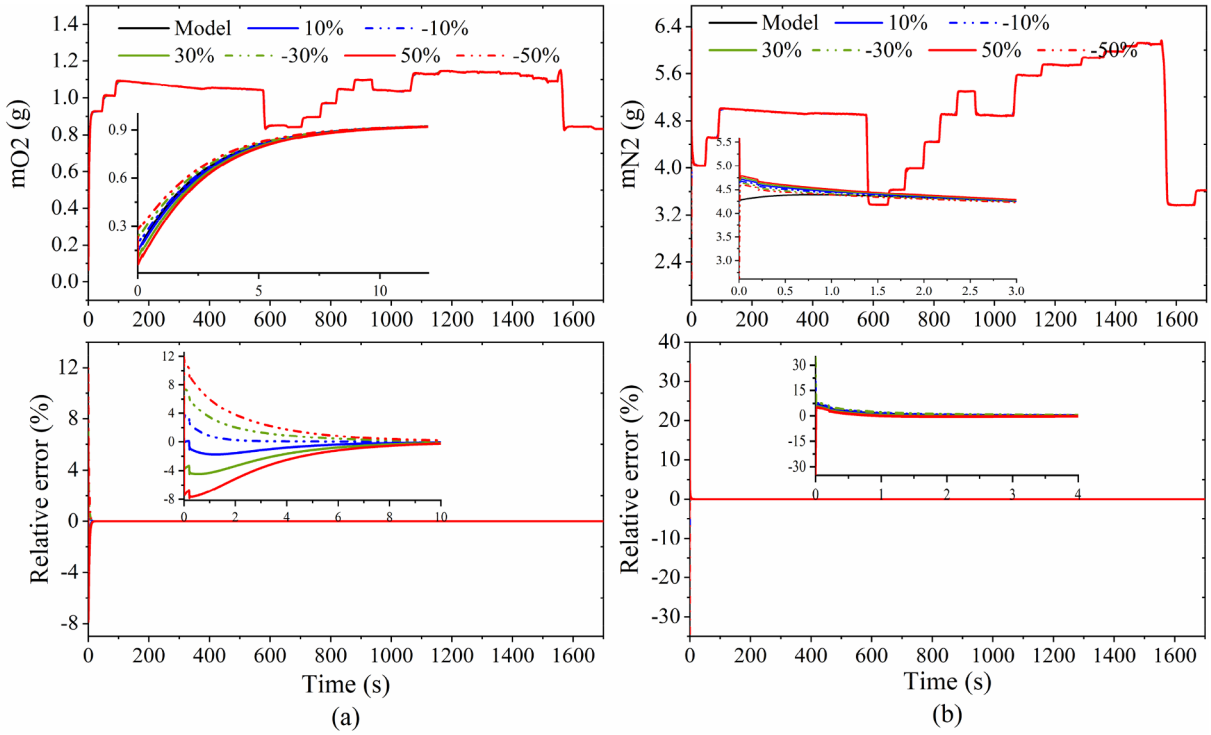


Fig. 10. Comparison of SVD-ACKF with different initial values; (a) estimation of oxygen mass; (b) estimation of nitrogen mass.

Table 6
Variations of system parameters.

Parameter	Nominal	Variation
V_{sm}	0.0057 m^3	+10%
V_{rm}	0.0043 m^3	+10%
$k_{sm,out}$	6.2194×10^{-6}	+5%
$k_{ca,out}$	3.8846×10^{-6}	+5%
T_{st}	348 K	+5%
T_{st}	348 K	+10%

of 0.1 ms, while the sampling frequency of the actual controller is ordinarily only about 100–150 Hz. Also, the model or algorithm should not be overly complicated in pursue of an accurate estimation of the observer, which will increase the operational cost of the controller and may be challenging to embed in the controller in the form of C code.

Another critical point is that offline values obtained based on models or fuel cell simulator are overwhelmingly applied for observer validation. As the on-line in-situ measurement technology of the internal state of fuel cells is still in the development stage, the rationality of the observation results can only be evaluated by the EIS and PCB measurement technology at this stage. Finally, it should be noted that although the observer may be able to accurately estimate the internal state within the fuel cell in the laboratory environment, the ambient noise is much higher in the real vehicle applications than in the laboratory. Moreover, when the battery pack is coupled with fuel cell systems to provide power, new interference problems may occur. Therefore, the parameters of the observer and model are demanded to debugged and calibrated continuously, and the commercialization of observer technology can only be carried out after repeated experiments and arguments.

6. Conclusions and future research orientations

This paper proposed a nonlinear internal state observer to estimate the mass of oxygen and nitrogen inside the cathode of a PEMFC system. A lumped parameters dynamic model was founded to describe the mass transport process on the cathode side. Based on this, a nonlinear observer was developed according to the ACKF algorithm with covariance matching. To improve the convergence of the ACKF observer, the singular value decomposition (SVD) was introduced to replace the Cholesky decomposition of state variables covariance. With a working dataset, the system model was parameterized using a nonlinear least-squares algorithm. The performance and effectiveness of the SVD-ACKF observer are validated and compared with the UKF observer and SVD-CKF observer with no adaptive process. Furthermore, the robustness against changes in the initial values of the states and model parameters, as well as different power requirements, are also tested. The results showed that such an SVD-ACKF observer demonstrated best convergence and accuracy, although a range of typical uncertainties exists. However, the relative error of stack temperature has a significant influence on the robustness of the observer.

There are several trends to be treated in future studies:

(1) Developing a nonlinear model for multi-state cooperative estimation with a comprehensive and accurate dynamic PEMFC system model is a significant trend.
(2) The simplified distributed parameter model which is more complex to provide more detailed information about the fuel cell in one direction, but also satisfy the demands of the system-level application, can be considered to acquire spatial details of internal states.
(3) The applicability of estimated states in a feedback control loop is required to be confirmed with different control algorithms. Besides, different kinds of observers such as Kalman filter observer and

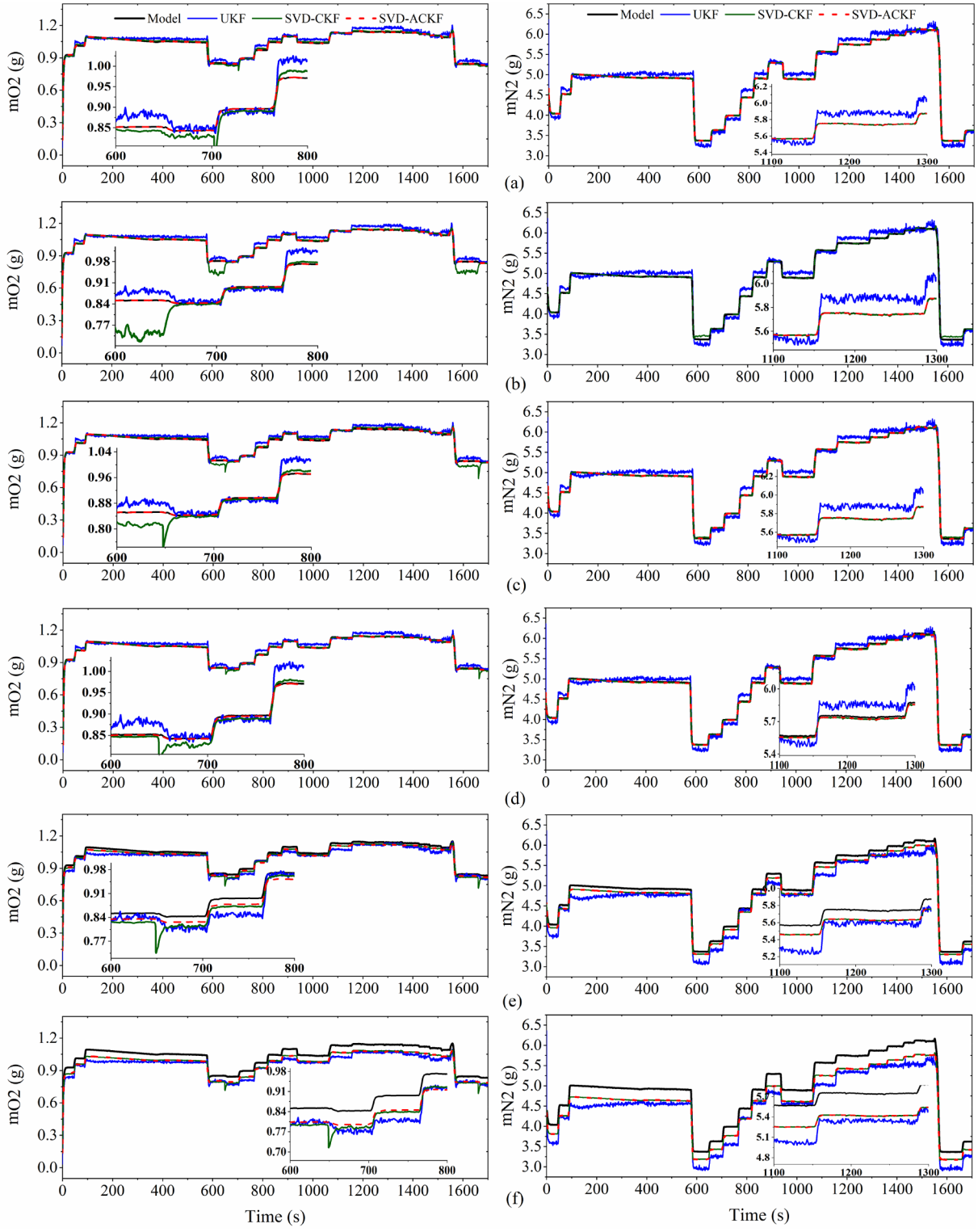


Fig. 11. Robustness of the observer against the uncertainties of different parameters; (a) V_{sm} + 10%; (b) V_{rm} + 10%; (c) $K_{sm,out}$ + 5%; (d) $K_{ca,out}$ + 5%; (e) T_{st} + 5%; (f) T_{st} + 10%.

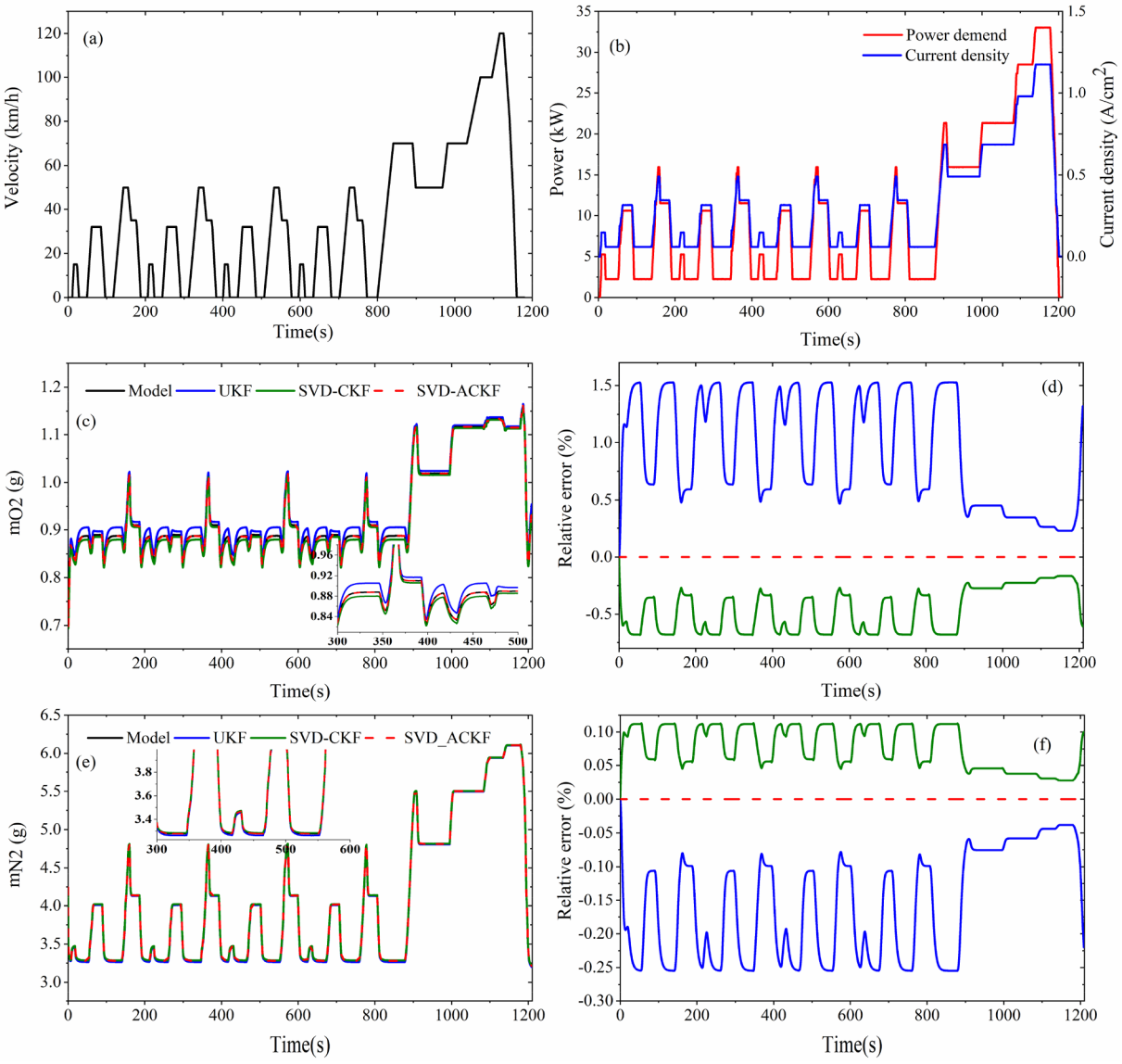


Fig. 12. Robustness of the observer against different power requirements; (a) NEDC working profile; (b) modified NEDC working profile used for fuel cell system test; (c) estimation of oxygen mass; (d) relative error of oxygen estimation; (e) estimation of nitrogen mass; (f) relative error of nitrogen estimation.

HOSM observer are required to be systematic compared to analyze the characteristics such as convergence speed, estimation accuracy, and calculation time.

(4) Experimental technologies like EIS and PCB measuring even innovative in-situ on-line measurement method, are required further to verify the rationality of the observer estimation results.

CRediT authorship contribution statement

Hao Yuan: Conceptualization, Methodology, Software, Data curation, Formal analysis. **Haifeng Dai:** Investigation, Formal analysis, Writing - original draft, Writing - review & editing. **Xuezhe Wei:** Investigation, Resources. **Pingwen Ming:** Visualization, Supervision.

Declaration of Competing Interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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Appendix A

$$f_1(x,u) = \frac{\eta_{cm}K_t}{R_{cm}J_{cp}}(c_1u_1 + c_2u_1^2 + c_3u_1^3 + c_4u_1^4 + c_5 - K_vx_1) - \frac{C_pT_{atm}}{x_1J_{cp}\eta_{cp}}\left[\left(\frac{x_2}{p_{atm}}\right)^{\frac{\gamma-1}{\gamma}} - 1\right]\cdot(B_{00} + B_{10}x_1 + B_{20}x_1^2 + B_{01}x_2 + B_{11}x_1x_2 + B_{02}x_2^2)$$

$$f_2(x,u) = \frac{\gamma R_a}{V_{sm}}\cdot(B_{00} + B_{10}x_1 + B_{20}x_1^2 + B_{01}x_2 + B_{11}x_1x_2 + B_{02}x_2^2)\cdot\frac{T_{atm}}{\eta_{cp}}\left(\eta_{cp} + \left(\frac{x_2}{p_{atm}}\right)^{\frac{\gamma-1}{\gamma}} - 1\right) - \frac{\gamma R_a}{V_{sm}}\cdot K_{sm,out}\left(x_2 - \frac{R_{O_2}T_{st}x_3}{V_{ca}} - \frac{R_{N_2}T_{st}x_4}{V_{ca}} - \phi_{ca}\cdot p_{sat}(T_{st})\right)u_3$$

$$f_3(x,u) = \frac{X_{O2,ca,in}M_{O_2}(x_2 - u_2p_{sat}(u_3))}{M_a(x_2 - u_2p_{sat}(u_3)) + M_vu_2p_{sat}(u_3)}\cdot k_{sm,out}\left(x_2 - \frac{R_{O_2}T_{st}x_3}{V_{ca}} - \frac{R_{N_2}T_{st}x_4}{V_{ca}} - \phi_{ca}\cdot p_{sat}(T_{st})\right) - \frac{x_3}{x_3 + x_4 + \frac{\phi_{ca}p_{sat}(T_{st})V_{ca}}{R_vT_{st}}}k_{ca,out}\left(\phi_{ca}p_{sat}(T_{st}) + \frac{R_{O_2}T_{st}x_3}{V_{ca}} + \frac{R_{N_2}T_{st}x_4}{V_{ca}} - x_5\right) - N\frac{M_{O_2}u_1}{4F}$$

$$f_4(x,u) = \frac{(1 - X_{O_2,ca,in})M_{N_2}(x_2 - u_2p_{sat}(u_3))}{M_a(x_2 - u_2p_{sat}(u_3)) + M_vu_2p_{sat}(u_3)}\cdot k_{sm,out}\left(x_2 - \frac{R_{O_2}T_{st}x_3}{V_{ca}} - \frac{R_{N_2}T_{st}x_4}{V_{ca}} - \phi_{ca}\cdot p_{sat}(T_{st})\right) - \frac{x_4}{x_3 + x_4 + \frac{\phi_{ca}p_{sat}(T_{st})V_{ca}}{R_vT_{st}}}k_{ca,out}\left(\phi_{ca}p_{sat}(T_{st}) + \frac{R_{O_2}T_{st}x_3}{V_{ca}} + \frac{R_{N_2}T_{st}x_4}{V_{ca}} - x_5\right)$$

$$f_5(x,u) = \frac{R_aT_{rm}}{V_{rm}}k_{ca,out}\left(\phi_{ca}p_{sat}(T_{st}) + \frac{R_{O_2}T_{st}x_3}{V_{ca}} + \frac{R_{N_2}T_{st}x_4}{V_{ca}} - x_5\right) - \frac{R_aT_{rm}}{V_{rm}}\frac{C_{rm}(\zeta_1u_1 + \zeta_2u_1^2 + \zeta_3u_1^3 + \zeta_4u_1^4 + \zeta_5u_1^5 + \zeta_6)(x_5 - p_{atm})}{\sqrt{Ru_4}}\sqrt{\gamma\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}}$$

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